RIVER VALLEY HIGH SCHOOL
JC 2
PRELIMINARY EXAMINATION

CANDIDATE NAME											
CENTRE NUMBER	S					CLASS	20J	INDEX NUMBER			

BIOLOGY **9744/03**

Paper 3 Long Structured and Free-response Questions **16 September 2021**

2 hours

Candidates answer on the Question Paper.
No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your Centre number, index number and name in the spaces at the top of this page.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Section A
Answer **all** questions in the spaces provided on the Question Paper.

Section B
Answer any **one** question in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate. You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together. The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
Section A	
1	
2	
3	
Section B	
Total	

This document consists of XX printed pages and X blank page.

Section A

Answer **all** the questions in this section.

1 **(a)** Discuss if cancer is a genetic disease. [3]

Cancer is a genetic disease

1. Result from accumulation of mutations;
2. Such as gain of function of proto-oncogene;
3. And loss of function mutation of tumour suppressor gene;
4. Which can be pass down to the next generation/inherited by offspring;
5. If the mutation occurs in germline cells;

Cancer may not be heritable

6. Due to mutation in somatic cells;
7. Offspring of cancer individual may not have cancer;

AO B

The transition of normal cells into the cancerous state is termed as cellular transformation and is accompanied by alteration in cell morphology as well as function. Fig 1.1 shows a group of cells, including cancer cells which has undergone cellular transformation.

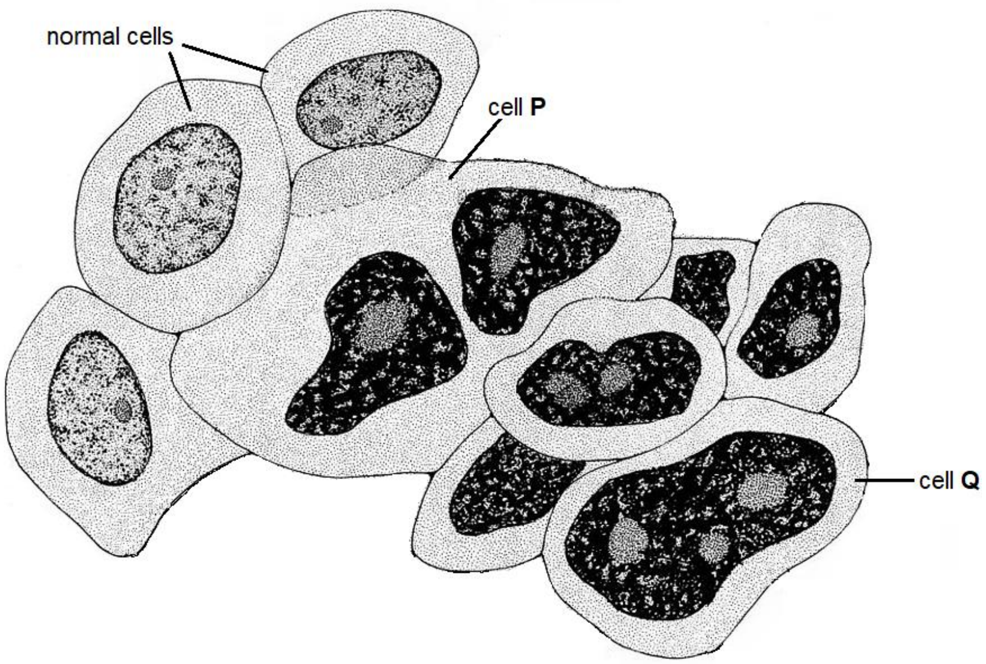


Fig. 1.1

(b) Apart from cell shape and size, identify another structural abnormality visible in cell **Q**, and explain how this abnormality may contribute towards cancerous growth. [3]

Abnormality (-1/2 mark for wrong tense)	Contribution towards cancer growth
1. Multiple nucleoli;;	2. Increased production of rRNA; 3. more ribosomes;

AO B

	4. facilitating <u>increased rate</u> of translation / synthesis of proteins; 5. more organelles, proteins produced which promote cell division / for making components of more cells;
1. Enlarged nucleus;;	2. Increased DNA content/chromosomes/ multiple copies of proto-ongogene/oncogene/gene amplification of proto-oncogene/oncogene; 3. More template for transcription; 4. Facilitating the <u>increased rate</u> of transcription of genes; 5. more organelles, proteins produced which promote cell division / for making components of more cells;

(c) Cells with morphology similar to cell **P** are common in cancer. These cells arose as a result of the failure of the mitotic cell cycle.

State the stage in which the mitotic cell cycle failed and explain how this led to the production of cell **P**. [4]

1. Cytokinesis;; AO B

2. Interphase results in growth of cell / production organelles & proteins;
3. The cells underwent mitosis / telophase;
4. But fail to form a cleavage furrow;
5. to separate/divide the cell into two;

6. Resulting in two nuclei;
7. And increase in cell size;

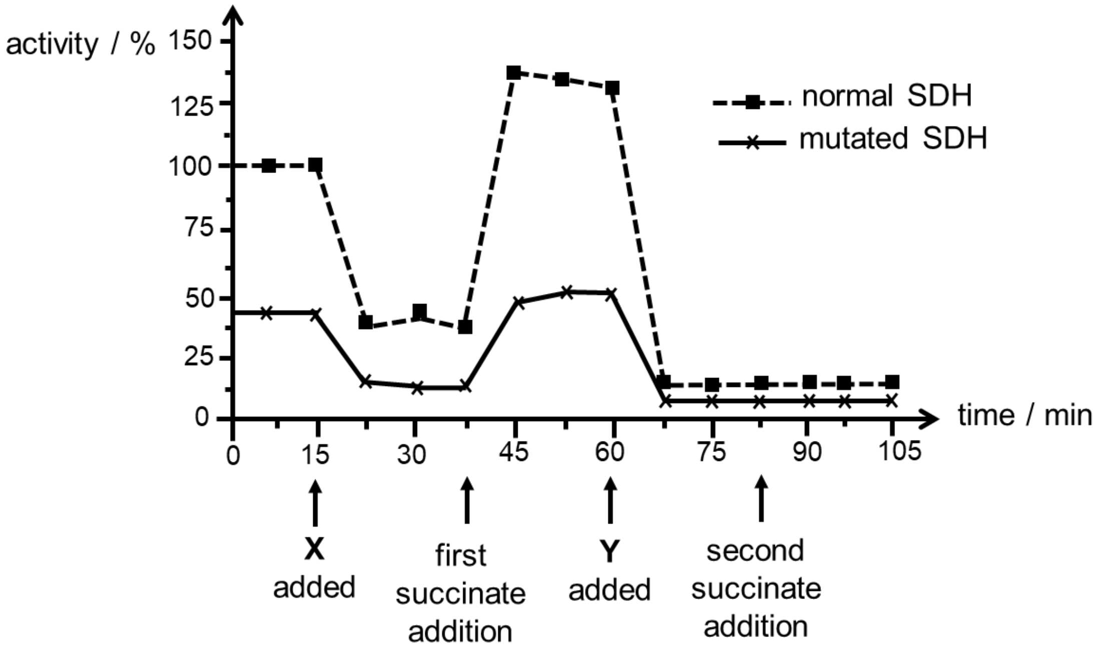
One feature of cancer cells is dysfunctional mitochondria, which is often due to mutations found in the genes coding for succinate dehydrogenase (SDH). SDH is a heterotetrameric protein complex, embedded on the inner mitochondrial membrane.

SDH is an important protein for both the Krebs cycle and oxidative phosphorylation. It catalyses the conversion of succinate to fumarate, in a reaction that is coupled with the reduction of FAD to $FADH_2$. SDH is also a member of the electron transport chain.

(d) Describe how one of the subunits of the SDH complex is synthesised from its mRNA. [5]

1. SDH mRNA serves the template;
2. mRNA molecule binds to the small ribosomal subunit (at the 5' end);
3. initiator tRNA binds to the start codon on the mRNA;
4. large ribosomal subunit binds to the small ribosomal subunit (forming a translational complex);
5. <u>anticodon</u> of an incoming tRNA molecule pairs with the codon at aminoacyl-tRNA site of ribosome (via complementary base pairing);
6. ribozyme/peptidyl transferase catalyse the peptide bond formation between amino acids;
7. ribosome translocates;
8. elongating polypeptide until ribosome reaches stop codon;
9. <u>release factor</u> binds directly to stop codon at A site;
10. polypeptide hydrolysed from the tRNA; AO
11. polypeptide folds into its three-dimensional shape; A

Inhibitors **X** and **Y** are known to affect SDH's role in the Krebs cycle. Fig. 1.2 shows changes to the activities of normal SDH and mutated SDH following the addition of inhibitor **X**, inhibitor **Y**, and an excess of succinate at different time points.



Adapted from https://cancerandmetabolism.biomedcentral.com/articles/10.1186/2049-3002-2-21

Fig. 1.2

(e) With reference to Fig. 1.2,

(i) account for the changes in activity of normal SDH following the addition of inhibitor **Y** and the second addition of succinate, [4]

1. Addition of inhibitor Y cause the activity of normal SDH to drop rapidly from 135 % to 12.5% in the first 7.5 minutes (accept quote time);;
2. Inhibitor Y binds to SDH at site other than active site;
3. Causes a change to the shape of active site of SDH/active site no longer complementary to succinate;
4. Less enzyme substrate formed per unit time;

5. Activity remains constant at 12.5% with the addition of succinate;;
6. Inhibitor Y binds irreversibly;
7. Addition of excess substrate cannot overcome the inhibition;

(ii) deduce **one** similarity in the structure of mutated and normal SDH. Justify your answer. [3]

1. Both have the similar shape of active site;;
2. Both activity is reduced following the addition of a competitive inhibitor X;;
3. Implying that the shape of competitive inhibitor is complementary to active site of both normal and mutated SDH; AO
4. Addition of excess succinate overcome the effect of molecule X; B

SDH is encoded by genes found in the nuclear DNA while some of the other electron carriers in the mitochondria are encoded by genes found in the mitochondrial DNA (mtDNA).

(f) Apart from size, describe how nuclear DNA is structurally different from mtDNA. [3]

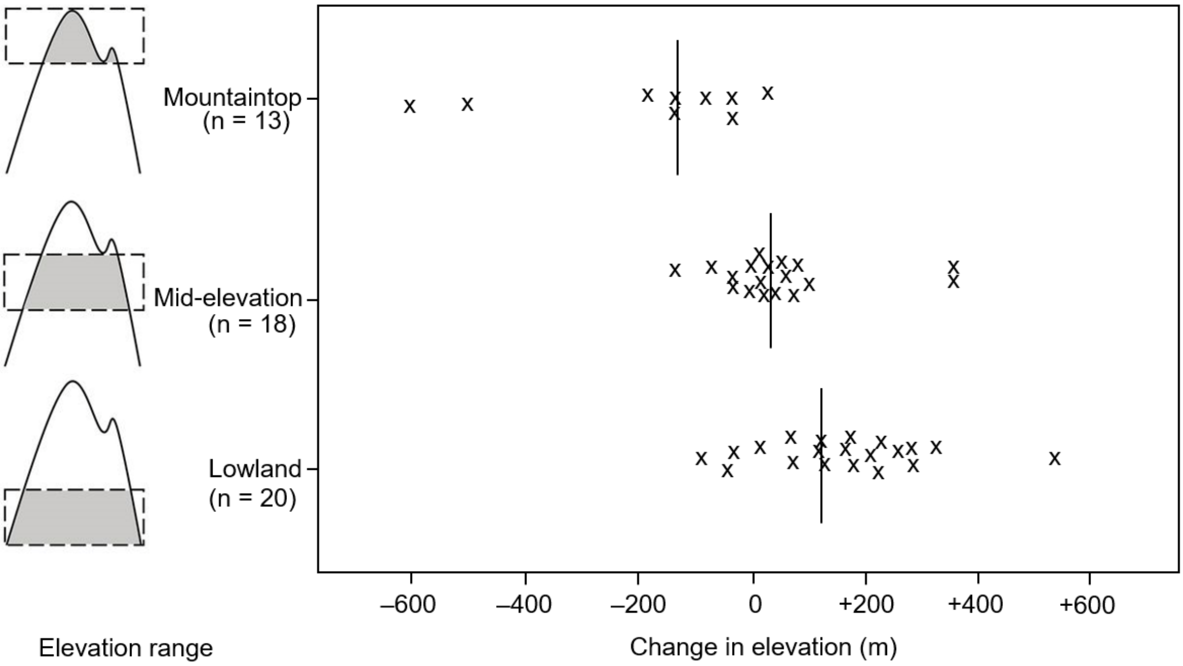
Features	Nuclear DNA	mtDNA
1. Shape	Linear	Circular
2. Telomere	Present	Absent
3. Centromere	Present	Absent
4. Introns	Present	Absent

AOA

[Total: 25]

2 Researchers followed 51 species in Malaysia over a 10-year period to assess the impact of increasing temperature on their habitat range on the mountain. Different species respond to increasing temperature differently, including migrating to a different elevation on the mountain.

Fig. 2.1 shows the change in elevation of each species' habitat at the different elevation range on the mountain after the 10-year study.



key

n = number of species originally present at that elevation range

x = data point for change in elevation for one species after the 10-year study

I = the average change at that elevation range

Fig. 2.1

Modified from Belshaw et al. Journal of Virology.Vol.84 2010

(a) With reference to Fig. 2.1, describe **two** differences in how species in different elevation ranges have responded to the increasing temperature. [2]

1. More species migrate to higher elevations in mid-elevation and lowlands than on mountaintop;
2. Quote values;
3. More species disappear / absent from survey on mountaintop than both mid-elevation and lowlands;
4. Quote values;
5. More species migrate to lower elevations on mountaintop than both mid-elevation and lowlands;
6. Quote values;
7. Extent of migration to lower elevations on mountaintop is larger than both mid-elevation and lowlands / lowland to higher elevation is larger than both mid-elevation and mountaintop;
8. Quote values;

AOB

9. Average change in elevation is negative on the mountaintop while that of both mid-elevation and lowlands is positive;
10. Quote values;
11. AVP;;

Any two

Genetic variation in a species is key to its survival in changing environments.

Fig. 2.2 shows the relationship between population size, geographical distribution and the amount of variation present in a species.

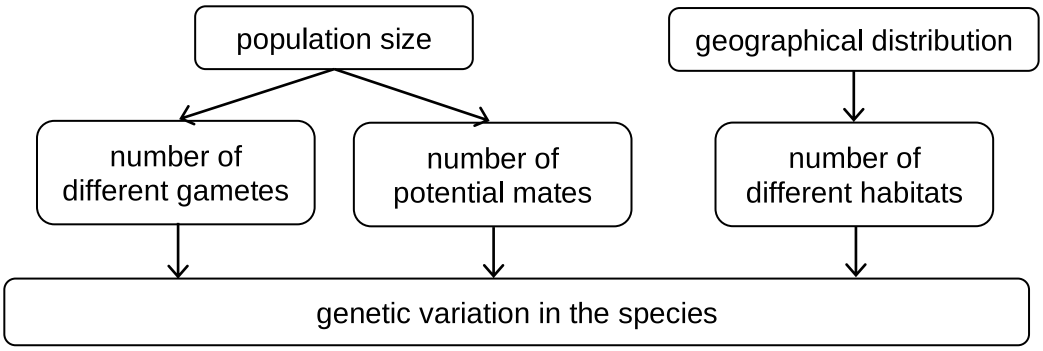


Fig. 2.2

Modified from Lougheed et al. Molecular Ecology.17(5) March 2008

(b) With reference to Fig. 2.2, explain how genetic variation is affected by an increase in population size and geographical distribution of a species. [4]

Population size

1. Increasing population size increases number of different gametes;
2. Due to an increased number of mutations in the population;
3. Increases number of new alleles;
4. Due to crossing over of non-sister chromatids (of homologous chromosomes) / independent assortment of bivalents;
5. which increases number of new combination of parental chromosomes / allelic combinations;

[Max. 2m]

6. Increased population size increases number of potential mates;
7. More possible combinations of zygotes due to random fertilisation;

Geographical distribution

8. Increasing geographical distribution increases number of different habitats / ecological niches;
9. Which increases types of selection pressure for populations of the species;
10. Resulting in increased number of different alleles in the gene pool of a species;

AOB

Unlike animals, plants have a more limited migration range. Therefore plants often have to adapt to changes in their habitat's environment.

The gene coding for aquaporin is highly varied in plant species.

Molecular and phylogenic analyses have traced the ancestral aquaporin gene to an early plant species. However, studies identified copies of the aquaporin gene present on different chromosomes in the genome of a descendant plant species.

(c) Suggest how mutations may have resulted in the distribution of aquaporin genes in the genome of the descendant plant species. [2]

1. Duplication of ancestral aquaporin gene;; or
2. Non-disjunction / failure to separate chromosomes during meiosis;;
   Results in multiple copies of aquaporin gene

3. Translocation of duplicated gene to another chromosome;; or
4. (Double-strand) breakage and re-joining of DNA of one chromosome with another;;

Max. 2m AOB

Aquaporins are integral membrane channel proteins. They play an important role in regulating the selectivity and permeability of water molecule movement across membranes.

Fig. 2.3 shows the structure of aquaporin. Aquaporin contains five loops that join transmembrane domains I to VI. In the primary sequence of aquaporin, the amino acid sequence asparagine-proline-alanine, also known as NPA, is highly conserved in the B and E loop.

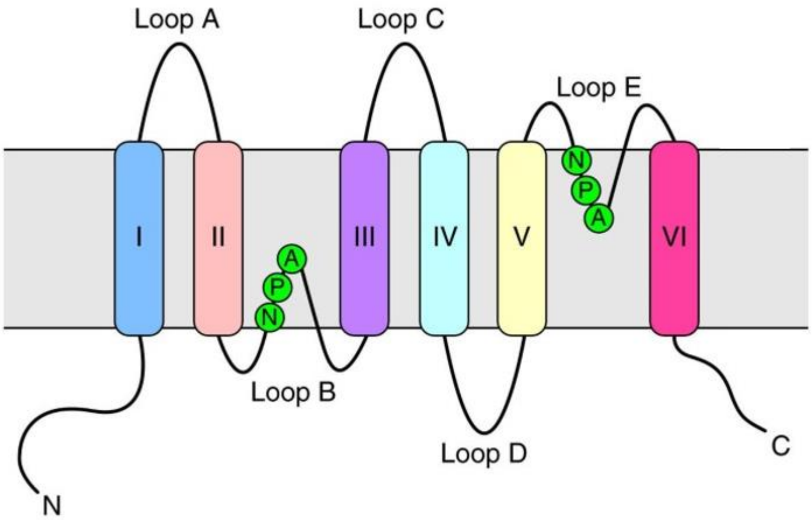


Fig. 2.3

The NPA sequences are present in the narrow central constriction of the channel as shown in Fig. 2.4.

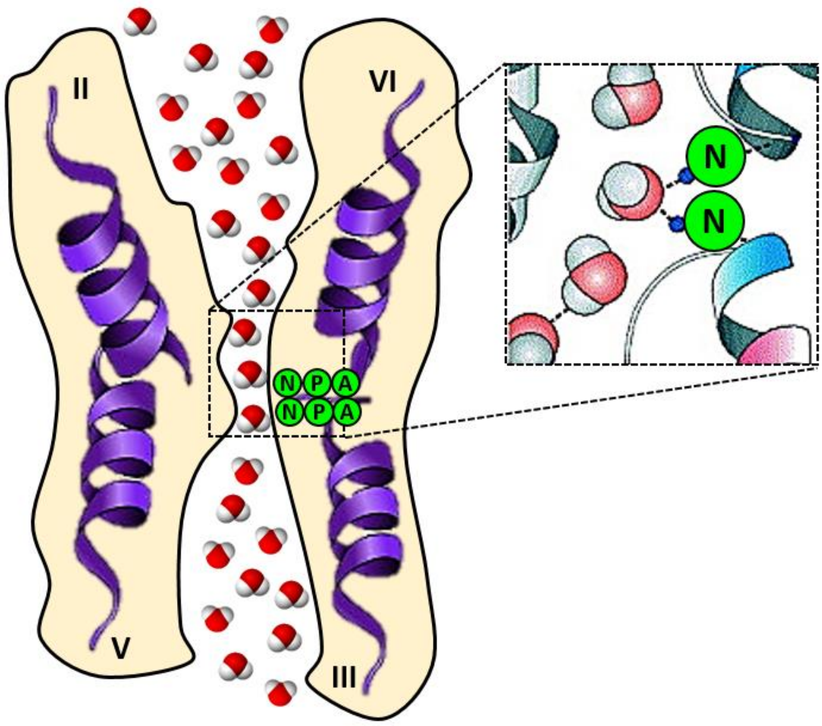


Fig. 2.4

Adapted from: Kruse *et al*. The aquaporins. Genome Biol 7, 206 (2006)

Fig. 2.5 shows the structure of NPA.

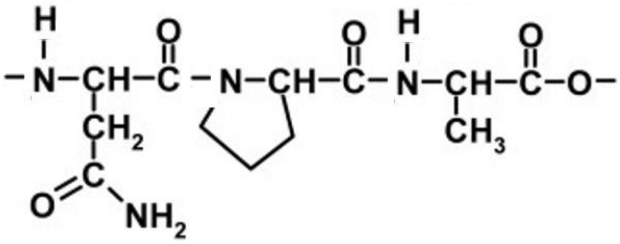


Fig. 2.5

(d) **(i)** Outline how the NPA sequences come together in the aquaporin channel. [2]

1. Folding of aquaporin polypeptide chain to give rise to α-helices (and β-pleated sheets);
2. Maintained by hydrogen bonding between C=O and N-H of polypeptide backbone;
3. Further folding into tertiary structure to bring NPA sequences close together in channel;
4. Maintained by R-group interactions (hydrogen bonds, ionic bonds, disulfide bridges and hydrophobic interactions); AOA

(ii) Explain how NPA regulates the selectivity of aquaporin to water molecules. [3]

1. Arrangement of NPA and helices II and V;
2. results in complementary shape;
3. and size of channel constriction to water molecules;
4. resulting in movement of water molecules in single file
5. Asparagine has polar R group to interact with water molecules;; AOB

(e) At higher temperatures, both membrane fluidity and permeability increase.

Explain how plants adapt to higher temperatures to reduce membrane fluidity and permeability. [2]

To reduce membrane fluidity

1. Increase length of hydrocarbon chains / FA tails of phospholipids;
2. Which increases surface area of contact / hydrophobic interactions between chains;
   OR
3. Increase degree of saturation / fewer carbon-carbon double bonds of hydrocarbon chains / FA tails of phospholipids;
4. Chains are more closely packed;

Max. 1m

To reduce membrane permeability

5. Presence of sterols / cholesterol-like compounds;
6. Which limits permeability of membrane; AOA

[Total: 15]
AOA – 4
AOB – 11

3 Embryonic stem cells (ESCs) are a potential source of medical therapy. To overcome ethical issues related to the use of ESCs, embryonic-like pluripotent stem cells can be generated from somatic cells by using either somatic cell nuclear transfer (SCNT) or by inducing formation of pluripotent stem cells (iPSCs). Fig. 3.1 outlines the two approaches.

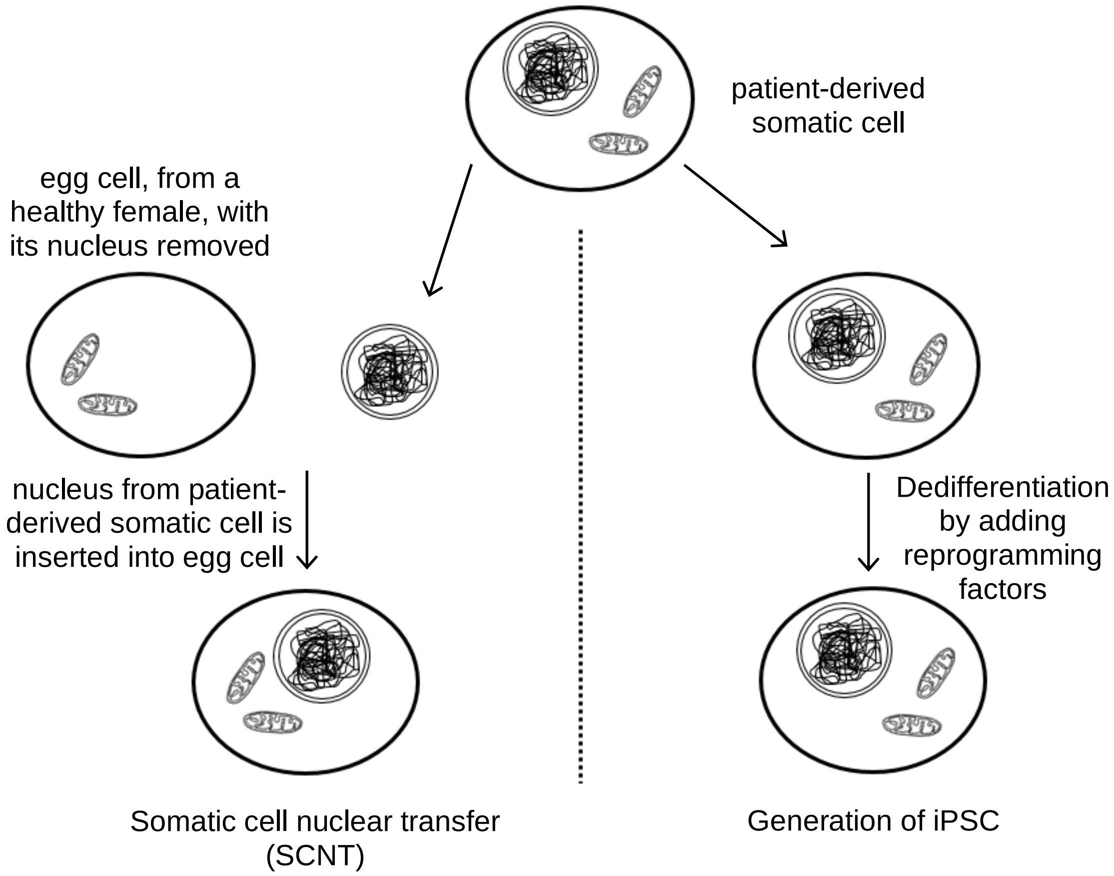


Fig 3.1

(a) **(i)** Describe the normal role of embryonic stem cells. [2]

1. Differentiate into almost any cell type;
2. To form any organ;
3. Except placenta;
4. Formation of three germ layers / endoderm, mesoderm and ectoderm; AO A

(ii) Using your knowledge and the information provided, fill in the table below by suggesting how the use of iPSCs for therapy has an advantage or disadvantage compared with the other stem cell types. [2]

Table 3.1

	iPSCs	
compared with	advantage	disadvantage
ESCs		1. Accumulated mutations in somatic cell will be present in iPSCs;; 2. AVP;;
SCNT-derived stem cells	1. Genetically-identical to patient somatic cell including mitochondrial DNA;; 2. AVP;;	

Max. 1m each AO B

In normal development, maternal mRNAs play an important role in the early development of an embryo.

- Maternal mRNAs are synthesised by transcription but are translationally repressed during the formation of oocytes. After entering the cytoplasm, maternal mRNAs immediately undergo deadenylation, are bound by regulatory proteins at the 5' UTR region and organised into membrane-less structures known as cytoplasmic granules.
- After fertilisation between the oocyte and sperm, a period of transcriptional silence occurs. During this period, development is driven by maternal mRNAs deposited in the zygote by the oocyte.
- Zygotic genome becomes transcriptionally active as maternal mRNAs are degraded. This is known as maternal to zygotic transition (MZT).

(b) **(i)** Explain how maternal mRNAs are translationally repressed. [2]

Deadenylation AO A

1. Decreased mRNA stability;
2. as poly(A)-binding protein unable to bind to shortened poly(A) tail; & AO

Regulatory proteins bound at 5' UTR B

3. Prevent access to translation initiation factors;
4. and binding of ribosome to 5' end of mRNA;

(ii) Suggest how the organisation of maternal mRNAs in cytoplasmic granules allow them to be stored. [1]

Aggregated mRNA (in granules) preventing access of nucleases to breakdown mRNA;; AO B

During MZT, the activity of maternal mRNA leads to a positive feedback, in the activity of the zygotic mRNA, that results in its degradation.

Fig. 3.2 shows the changes in transcript levels of maternal mRNA and zygotic mRNA during MZT.

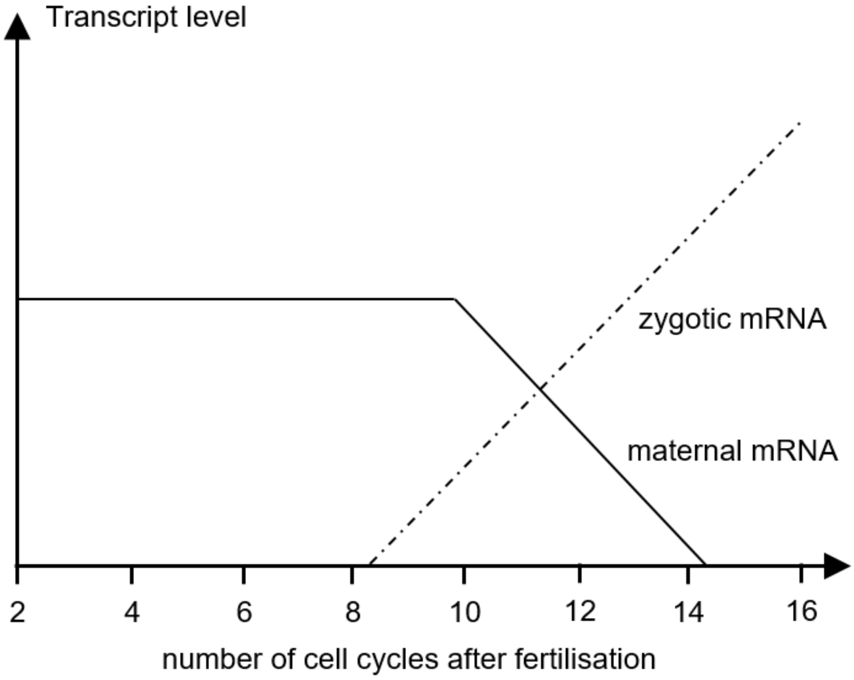


Fig. 3.2

Table 3.2 shows the changes in expression of selected enzymes that result in the change in transcript levels during MZT. Of these enzymes, only one is encoded by the maternal mRNA.

Table 3.2

	number of cell cycles from fertilisation							
enzymes	2	4	6	8	10	12	14	16
decapping enzymes				++	++	++	+	

DNA demethylation enzymes			+++	++	++	+	+	
nuclease					++	++	++	+

Key

+ low rate of expression

++ moderate rate of expression

+++ high rate of expression

(ii) With reference to Fig. 3.2 and Table 3.2, explain how expression of maternal mRNAs results in its degradation. [3]

1. Translation of maternal mRNAs;
2. To synthesise DNA demethylation enzymes (at 6^{th} cell cycle);

3. Demethylation of zygotic DNA;
4. RNA polymerase able to bind to promoter;
5. Leads to expression of decapping enzymes and nucleases (at 8^{th} cell and 10^{th} cell cycle);

6. Decapping enzymes remove 5’ cap of maternal mRNAs;
7. Nucleases cleave/breakdown maternal mRNA;

Max. 3m

AO B

[Total: 10]

Section B

5 **(a)** Outline how glucagon in the bloodstream returns blood glucose concentration back to set point and describe the advantages of cell signaling pathways in multicellular organisms. [15]

Regulation of blood glucose concentration (10m)

1. Glucagon binds to binding site on GPLR on target cells;
2. Conformation change to activate GPLR;
3. Binds to inactive G protein;
4. Activates G protein by displacing GDP with GTP;
5. Activated G protein binds adenyl cyclase;
6. Converts ATP to cAMP (second messenger);
7. Which activates downstream proteins / phosphorylation cascade;
8. Activates transcription factors to initiate expression of genes;

9. In liver cells;
10. Activation of enzymes involved in glycogenolysis;
11. Where stored glycogen is converted to glucose;

12. Inhibition of enzymes involved in glycogenesis;
13. Prevent glucose from being converted to glycogen;

14. Activation of enzymes involved in gluconeogenesis;
15. Absorption of amino acids, fatty acids and glycerol from bloodstream;
16. And converted to glucose;

17. In adipose tissues;
18. Activation of enzymes involved in lipolysis;
19. To breakdown triglycerides into fatty acids and glycerol;

20. Thereby raising the blood glucose concentration back to set point;

Advantages (4m)

Signal amplification:

16. A single signal molecule can result in a large number of activated molecules at the end of the pathway;;

Multistep pathways:

17. multistep pathways afford greater opportunity for coordination and regulation allowing for fine-tuning of cellular response;;

Relay system:

18. Relay system allows for binding of receptors by signals to activate genes in the nucleus;;

Numerous reactions:

19. A single ligand can trigger numerous reactions at once within a cell;;

Simultaneous activation of many cells:

20. A single type of ligand can simultaneously activate many cells;;

Max. 4m for listing of advantages

QWC: Answers correctly include reception, signal transduction and cellular response and at least 2 advantages;;

(b) Explain how features of proteins make them suited for their roles in cell signaling pathways. [10]

Specificity:
1. Folding of polypeptide into specific 3-dimensional conformation;
2. Results in specificity of active site / binding site;
3. Complementary shape to substrate / ligand;
4. Complementary charge;

Diversity:
5. 20 different types of (naturally-occurring) amino acids;
6. Joined together in different sequences;
7. And of different lengths;
8. Results in huge diversity in shapes / structures / 3D conformation;
9. May be further assembled into quaternary structures;
10. May be bound to / require cofactors to function;
11. May be embedded in membrane;
12. Due to non-polar R-groups of amino acids on exterior of protein;
13. Or present in cytoplasm / soluble in aq. medium;
14. Due to polar R-groups of amino acids on exterior of protein / globular protein;
15. May provide hydrophilic channels;
16. Due to polar & charged R-groups of amino acids on interior of protein;

Activity:
17. R-group interactions of amino acid residues;
18. Allows interaction with diverse biomolecules / DNA, proteins;
19. Catalytic residues to catalyse enzymatic reactions;
20. Contact residues to bind to target molecules;
21. Can be activated/inactivated easily by biochemical modifications;
22. Ability to change conformation (for activation);
23. Due to weak interactions within / between subunits;

^^^Students may interpret "features" as structural features instead, resulting in different answer format.

QWC: Well-structured and relate correctly to at least 2 different features;;

[Total: 25]

6 **(a)** Using a named example, describe how a bacteria can cause human diseases and discuss how penicillin can eliminate gram-negative bacteria. [15]

1. A person inhales droplet nuclei containing *M. tuberculosis*;;
2. the droplet nuclei traverse the mouth or nasal passages, upper respiratory tract, and bronchi to reach the alveoli of the lungs;
3. These bacilli are ingested by alveolar macrophages;
4. but some escape the effects of alveolar macrophages/some are not digested;
5. and multiply in macrophages;
6. leading to the destruction of alveolar macrophages.;
7. Antigen presenting dendritic cells travelled to the lymph nodes;
8. where T lymphocytes are activated and recruited. ;
9. Activated T lymphocytes that migrate to the site of infection;
10. proliferate and surround the tubercle bacilli;
11. forming a granuloma, ;
12. where macrophages become activated to kill intracellular M. tuberculosis.;
13. The formation of granulomas marks the latent stage of infection.;
14. Reactivation occurs;
15. when there is decline in the host's immunity due to genetic or environmental cause;
16. the granuloma structure disrupts;
17. and results in pulmonary disease.;
18. these bacilli may spread by way of lymphatic channels or through the bloodstream to more distant tissues and organs;

19. Penicillin have chemical structure known as the β-lactams ring.;
20. The β-lactams ring shape mimics the D-Ala-D-Ala peptide terminus;
21. that serves as the natural substrate for the activity of Penicillin Binding Proteins (PBP) during cell wall synthesis.;
22. β-lactams ring of antibiotics binds PBPs,
23. rendering them unable to perform their role in cell wall synthesis.;
24. Penicillin interferes with interpeptide linking of peptidoglycan.;
25. Cell walls without intact peptidoglycan cross-links are structurally weak;
26. prone to collapse and disintegrate when the bacteria attempts to divide. ;
27. This then leads to death of the bacterial cell due to osmotic instability or autolysis. ;

QWC – identified named disease and describe both the disease and antibiotic mechanism.;;

(b) Explain how features of biological membranes facilitate immune response in humans. [10]

Structure of membrane

1. Biological membranes are made up of phospholipids;
2. Which arranges to form a lipid bilayer;
3. Held by weak hydrophobic interactions;
4. Contains cholesterol which prevents close packing
5. And phospholipids with unsaturated fatty acids
6. To increase/maintain fluidity of the membrane;

Fluidity:

endocytosis

7. Allowing the macrophages/dendritic cell/antigen presenting cells/B cells to undergo phagocytosis/endocytosis;
8. to engulf pathogens to remove them from the bloodstream
9. whereby cell surface membrane invaginate and pinches in to form an endocytic vesicle;

10. so that APC/B cell receptor can present antigen peptide to T cell on class II MHC;

formation of vesicle

11. Formation of vesicle containing pathogens in phagocytes;
12. To contain/isolate pathogens in in vesicles protecting the cell;

fusion of vesicle

13. Lysosome fuses with phagocytic vesicle to digest the pathogen;

exocytosis

14. Allowing for exocytosis to occur;
15. as secretory vesicles can fuse with cell surface membrane;
16. whereby antibodies produces in plasma cells can be packaged into secretory vesicles to be secreted out of cell via exocytosis;
17. so that antibodies can bind to pathogens in blood;

18. Allow for the action of cytotoxic T cells to kill infected cell;
19. By embedding pore complex onto the cell surface membrane;

AVP

Mosaic:

20. Biological membranes are embedded with proteins;

recognition

21. Biological membrane contain such as receptor proteins
22. immune cells such have cell surface receptors that allows them to recognise pathogens
23. Quote example – BCR/TCR on B cell/T cell
24. which binds to pathogens via complementary shape to activate the cell;

25. allows them to receive chemical signals from other immune cells;
26. Quote example, receptor for cytokines/interleukin 1/interleukin 2 (on helper T lymphocyotes/ on B cells)

27. Biological membranes have glycolipids/glycoproteins;

identity

28. which acts as cell surface markers;
29. such as class I MHC which are form on all normal cells;
30. to differentiate between self and non-self to initiate immune response on non-self;

AVP

QWC: Student address both the fluid and mosaic nature of membrane and matches the properties/features to at least one feature of immune response